

Lecture 13

Metric, Normed, and Hilbert Spaces

Inner products are conjugate-linear in the first slot (physics convention): $\langle x, y \rangle = \sum_n \overline{x_n} y_n$ on ℓ^2 and $\langle f, g \rangle = \int \overline{f} g$ on L^2 . A Hilbert basis means a complete orthonormal family (countable only when separability is assumed), and $\mathcal{H}^*$ means the continuous dual; complex Riesz is conjugate-linear. A ★ marks a harder problem.

Core tutorial route

Work **13.2, 13.8, 13.11, 13.28, 13.27, 13.29, 13.21, and 13.32** in that order (about 85–100 minutes before discussion). This route covers metric versus norm and a parallelogram decision; a concrete incomplete space and completion; measure-aware ℓ^2/L^2 reasoning; basis completeness versus ambient completeness; the Bessel–expansion proof spine; closed-subspace best approximation; one justified Fourier bridge; and bounded complex Riesz.

Additional practice: 13.1, 13.3–13.7, 13.10, 13.12–13.19, 13.22–13.23, 13.26, 13.30–13.31, 13.33, and 13.35. **Bridge:** 13.20 and 13.24–13.25 (Fourier modes and approximation). **Extension:** 13.9 (absolute-series completeness) and 13.34 (evaluation spikes). Bridge and extension tasks do not count toward core progress.

Metrics, norms, completeness

Exercise 13.1 (proof) Show that $d(x, y) = \|x - y\|$ satisfies the triangle inequality whenever $\|\cdot\|$ is a norm.

Exercise 13.2 (proof) On $C[0, 1]$ let $\|f\|_\infty = \sup_x |f(x)|$ and $d(f, g) = \|f - g\|_\infty$. Explain why d is induced by a norm, then decide whether that norm comes from an inner product by testing $f = 1$ and $g(x) = x$.

Answer. d is the metric induced by the sup norm; the parallelogram law fails ($5 \neq 4$), so the norm is not inner-product induced.

Exercise 13.3 (computational) For $x = (2, -1, 2) \in \mathbb{R}^3$ compute $\|x\|_1$, $\|x\|_2$, and $\|x\|_\infty$.

Answer. $\|x\|_1 = 5$, $\|x\|_2 = 3$, $\|x\|_\infty = 2$.

Exercise 13.4 (computational) For $x = (1, -2, 2, -4) \in \mathbb{R}^4$ compute $\|x\|_1$, $\|x\|_2$, and $\|x\|_\infty$.

Answer. $\|x\|_1 = 9$, $\|x\|_2 = 5$, $\|x\|_\infty = 4$.

Exercise 13.5 (computational) For $x = (3i, -4) \in \mathbb{C}^2$ compute $\|x\|_1$, $\|x\|_2$, and $\|x\|_\infty$.

Answer. $\|x\|_1 = 7$, $\|x\|_2 = 5$, $\|x\|_\infty = 4$.

Exercise 13.6 (computational) Does the taxicab norm $\|\cdot\|_1$ on $\mathbb{R}^2$ come from an inner product? Test the parallelogram law with $u = (1, 0)$, $v = (0, 1)$.

Answer. No: $8 \neq 4$, so $\|\cdot\|_1$ comes from no inner product.

Exercise 13.7 (computational) Verify the parallelogram law for the Euclidean norm on $\mathbb{R}^2$ with $u = (3, 0)$, $v = (0, 4)$, consistent with $\|\cdot\|_2$ coming from an inner product.

Answer. Both sides equal 50; the law holds.

Exercise 13.8 (proof) Let q_n be the decimal truncation of π after n digits, regarded as an element of $\mathbb{Q}$ with the usual metric. Show that (q_n) is Cauchy in $\mathbb{Q}$ but does not converge there. Name its completion and its limit in that completion.

Answer. The tail is bounded by 10^{-n} , so (q_n) is Cauchy; it has no rational limit. The completion is $\mathbb{R}$, where $q_n \rightarrow \pi$.

Exercise 13.9 (★) **Optional extension.** Prove that a normed space is complete if and only if every absolutely convergent series converges.

Hint. For the forward direction, show that the partial sums are Cauchy. For the converse, extract from a Cauchy sequence a subsequence with successive gaps at most 2^{-k} , then telescope.

Exercise 13.10 (proof) Prove the reverse triangle inequality $|\|u\| - \|v\|| \leq \|u - v\|$ in any normed space.

ℓ^2 and L^2

Exercise 13.11 (proof) Answer both parts.

1. Is $x_n = 1/n$ in ℓ^2 ? Is $x_n = 1/\sqrt{n}$?
2. In $L^2[0, 1]$, let $f = 0$ and let g equal 1 at $x = 0$ and 0 elsewhere. Explain why f and g are the same L^2 vector and why this prevents point evaluation at 0 from being a functional on L^2 .

Answer. $(1/n) \in \ell^2$ and $(1/\sqrt{n}) \notin \ell^2$; $f = g$ almost everywhere, so they represent one class but have different point values.

Exercise 13.12 (computational) Compute $\langle f, g \rangle$ for $f(x) = 1$, $g(x) = x^2$ on $L^2[0, 1]$, and $\|g\|$.

Answer. $\langle f, g \rangle = \frac{1}{3}$, $\|g\| = 1/\sqrt{5}$.

Exercise 13.13 (computational) Normalize $\psi(x) = e^{-x}$ on $[0, \infty)$ in L^2 .

Answer. $\|\psi\| = 1/\sqrt{2}$; normalized state $\sqrt{2} e^{-x}$.

Exercise 13.14 (computational) For $x = (1, \frac{1}{2}, \frac{1}{4}, \dots) \in \ell^2$ (that is $x_n = 2^{-n}$, $n \geq 0$) compute $\|x\|$.

Answer. $\|x\| = 2/\sqrt{3}$.

Exercise 13.15 (computational) For $x_n = 2^{-n}$ and $y_n = 4^{-n}$ ($n \geq 0$) in ℓ^2 , compute $\langle x, y \rangle$.

Answer. $\langle x, y \rangle = \frac{8}{7}$.

Exercise 13.16 (computational) For which real r is $(1, r, r^2, r^3, \dots) \in \ell^2$, and what is its norm then?

Answer. $|r| < 1$; then $\|x\| = 1/\sqrt{1 - r^2}$.

Exercise 13.17 (computational) Compute $\langle f, g \rangle$ for $f(x) = x$, $g(x) = x^2$ on $L^2[0, 1]$.

Answer. $\langle f, g \rangle = \frac{1}{4}$.

Exercise 13.18 (computational) Normalize $\psi(x) = x^2$ on $[0, 1]$ in L^2 .

Answer. $\|\psi\| = 1/\sqrt{5}$; normalized state $\sqrt{5} x^2$.

Exercise 13.19 (proof) Prove that if $x = (x_n) \in \ell^2$ then $x_n \rightarrow 0$. Show by example that the converse fails.

Orthonormal bases and Fourier series

Exercise 13.20 (computational) The functions $\frac{1}{\sqrt{2\pi}}e^{inx}$ are orthonormal on $L^2[-\pi, \pi]$. Verify orthonormality for $n = 0$ and $n = 1$.

Answer. $\langle e_0, e_0 \rangle = \langle e_1, e_1 \rangle = 1$, $\langle e_0, e_1 \rangle = 0$.

Exercise 13.21 (proof) **Bridge task within the core route.** The L^2 Fourier calculation for x^2 on $[-\pi, \pi]$ gives

$$x^2 = \frac{\pi^2}{3} + 4 \sum_{n \geq 1} \frac{(-1)^n}{n^2} \cos(nx) \quad \text{in } L^2.$$

Explain why the series converges uniformly, why its continuous limit must equal x^2 at every point, and only then evaluate at 0 and π to find the alternating and ordinary sums of reciprocal squares.

Answer. The M-test applies since $4/n^2$ is summable; continuity plus a.e. equality upgrades the identity to every point. Thus $\sum (-1)^{n+1}/n^2 = \pi^2/12$ and $\sum 1/n^2 = \pi^2/6$.

Exercise 13.22 (computational) On $L^2[-1, 1]$ show that 1 and x are orthogonal, and compute $\|1\|$ and $\|x\|$.

Answer. $\langle 1, x \rangle = 0$; $\|1\| = \sqrt{2}$, $\|x\| = \sqrt{2/3}$.

Exercise 13.23 (computational) Normalize 1 and x on $L^2[-1, 1]$ to obtain the orthonormal pair e_0, e_1 .

Answer. $e_0 = 1/\sqrt{2}$, $e_1 = \sqrt{3/2} x$.

Exercise 13.24 (computational) Verify orthonormality of $e_n = \frac{1}{\sqrt{2\pi}}e^{inx}$ on $L^2[-\pi, \pi]$ for the indices $n = 2, 3$: compute $\langle e_2, e_2 \rangle$ and $\langle e_2, e_3 \rangle$.

Answer. $\langle e_2, e_2 \rangle = 1$, $\langle e_2, e_3 \rangle = 0$.

Exercise 13.25 (computational) Find the best approximation of $f(x) = x$ on $L^2[-\pi, \pi]$ by the orthogonal set $\{\sin x, \sin 2x\}$.

Answer. $2 \sin x - \sin 2x$.

Exercise 13.26 (computational) Let $\{e_1, e_2, e_3\}$ be orthonormal in a Hilbert space and v satisfy $\|v\|^2 = 10$ with $\langle e_1, v \rangle = 1$, $\langle e_2, v \rangle = 2$, $\langle e_3, v \rangle = 2$. Compute $\|Pv\|$, where P projects onto $\text{span}\{e_1, e_2, e_3\}$, and the squared distance $\|v - Pv\|^2$; check Bessel's inequality.

Answer. $\|Pv\| = 3$, $\|v - Pv\|^2 = 1$; Bessel holds since $9 \leq 10$.

Exercise 13.27 (proof) Let (e_n) be an orthonormal sequence in a Hilbert space and $s_N = \sum_{n \leq N} \langle e_n, v \rangle e_n$. Prove the expansion spine: finite Pythagoras gives Bessel; coefficient tails make (s_N) Cauchy; ambient Hilbert completeness gives a limit s ; and $v - s$ is orthogonal to the closed span. Conclude that Parseval equality is equivalent to completeness of the family.

Hint. First compute $\|v - s_N\|^2$. Then compute $\|s_M - s_N\|^2$ and test the limiting residual against each e_k .

Exercise 13.28 (proof) Separate these two meanings of “complete.” Let c_{00} be the finite-support sequences with the ℓ^2 norm, and let (e_n) be the standard coordinate family.

1. Show (e_n) has dense span in both c_{00} and ℓ^2 .
2. Explain why the family is complete in that basis sense in both spaces, while only the ambient space ℓ^2 is complete as a metric space.

Projection and Riesz

Exercise 13.29 (computational) Find the orthogonal projection of $f(x) = x$ onto the constants in $L^2[0, 1]$, and the residual.

Answer. $Pf = \frac{1}{2}$; residual $x - \frac{1}{2}$, orthogonal to the constants.

Exercise 13.30 (proof) Let $m(0) = 1$ and $m(x) = 0$ for $x \neq 0$ on $[0, 1]$. Find the pointwise supremum, essential supremum, and norm of the multiplication operator M_m on $L^2[0, 1]$. Explain what this tests.

Answer. $\sup |m| = 1$, $\text{ess sup } |m| = 0$, and $M_m = 0$, so $\|M_m\| = 0$.

Exercise 13.31 (computational) Find the orthogonal projection of $f(x) = x^3$ onto $\text{span}\{1, x\}$ in $L^2[-1, 1]$.

Answer. $Pf = \frac{3}{5}x$.

Exercise 13.32 (proof) On complex ℓ^2 define $\varphi(x) = \sum_{n \geq 1} ix_n/3^n$. Prove that it is bounded, find the Riesz vector u with $\varphi = \langle u, \cdot \rangle$, and verify directly that the Riesz map $R(u) = \langle u, \cdot \rangle$ satisfies $R(\alpha u) = \bar{\alpha}R(u)$.

Answer. $u_n = -i/3^n$; Cauchy–Schwarz proves boundedness; R is conjugate-linear.

Exercise 13.33 (computational) By Riesz, find $u \in L^2[0, 1]$ with $\varphi(f) = \langle u, f \rangle$ for the functional $\varphi(f) = \int_0^1 (2 + 3x) f(x) dx$.

Answer. $u(x) = 2 + 3x$.

Exercise 13.34 (★) **Optional extension.** On $C[0, 1]$ equipped with the L^2 norm, define $f_n(x) = \max(1 - nx, 0)$. Show that point evaluation $E(f) = f(0)$ is linear and well-defined there but unbounded. Explain why the same formula is not even well-defined on the quotient space $L^2[0, 1]$.

Hint. Compute $\|f_n\|_2^2$ and compare $E(f_n)$; then change one representative at the point 0.

Exercise 13.35 (proof) Show that the vector u representing a bounded functional via Riesz is unique: if $\langle u, \cdot \rangle = \langle u', \cdot \rangle$ as functionals, then $u = u'$.